变更历史：

变更日期	变更人	变更内容	备注
20241227	ID10086	Add pullup at NOVP and NOVPX Add AND on NFSS1 and DIAHVASC_Pxx Add AND on NFSS1 and DIAHVASCX_Pxx Add DAP Pin9 to TDI fot JTAG	
20250102	ID10086	Add GSENH GSENL on P13_0/1/2/3 Assign AN41 to chopper bus voltage Add chopper control on P01_/4/5/6/7 Delete SPI-Controlled LSD	
20250107	ID10086	Add 47Ohm current-limitor for logic output P40.6 Rename as TRES1, brake resistaor temperature1 P40.8 Rename as TRES2, brake resistaor temperature2	
20250108	ID10086	Update ICT point	
20250227	ID10086	Add HVIL detection	
20250303	ID10086	Add 50pin for inter-connection	
20250814	ID10086	Delete X1 BMS CAN interface	
20250828	ID10086	Delete reserved 0Ohm	
20250915	ID10086	Delete external 48pin info	
20250922	ID10086	Release A0	

For SMT

09:38:43 Monday, September 22, 2025

NO.	Revision	DSN.	Date		Name	Date
				DSN.		
				DWG.		
				CHK.		
				ADT.		

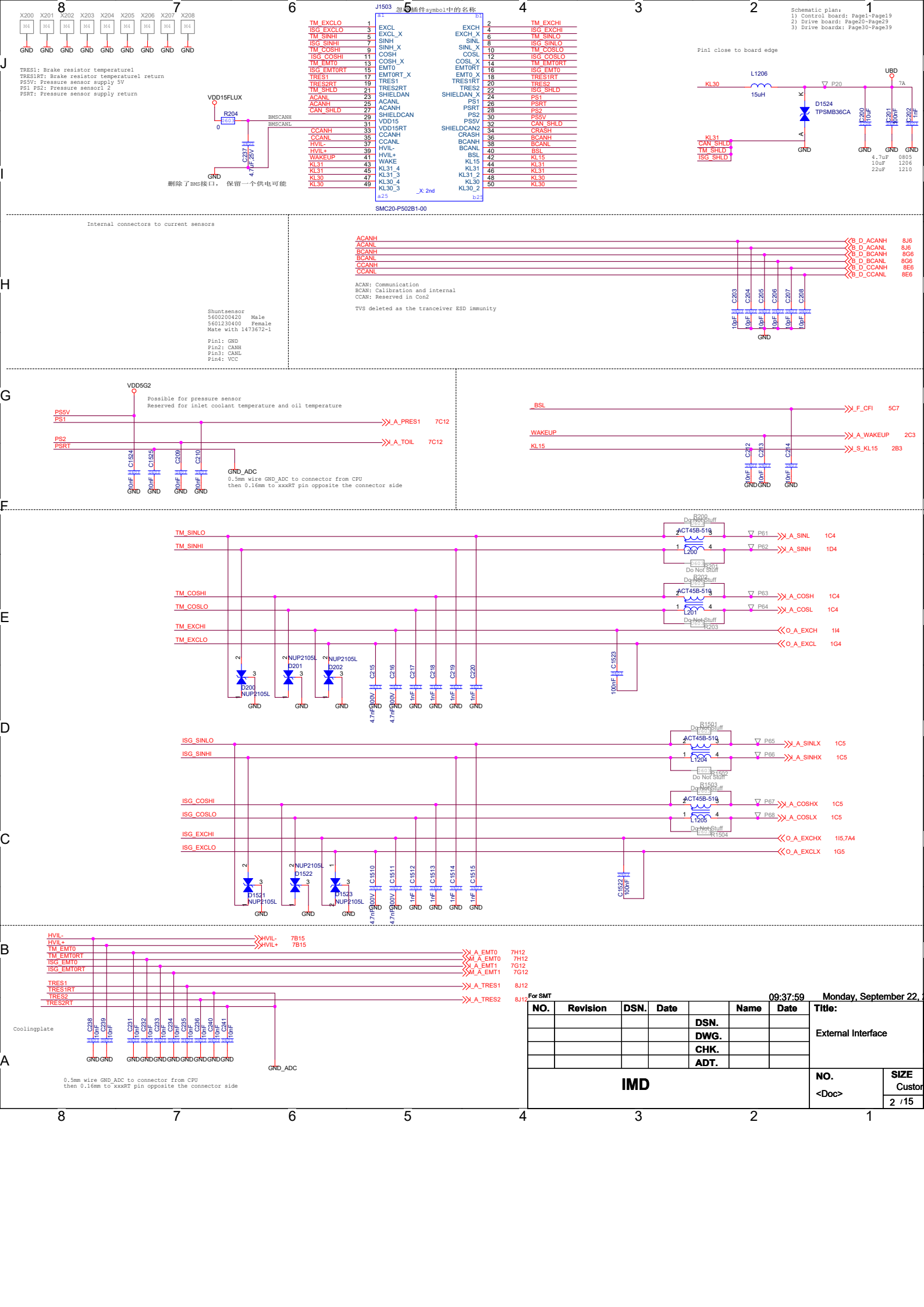
Title: Revision History

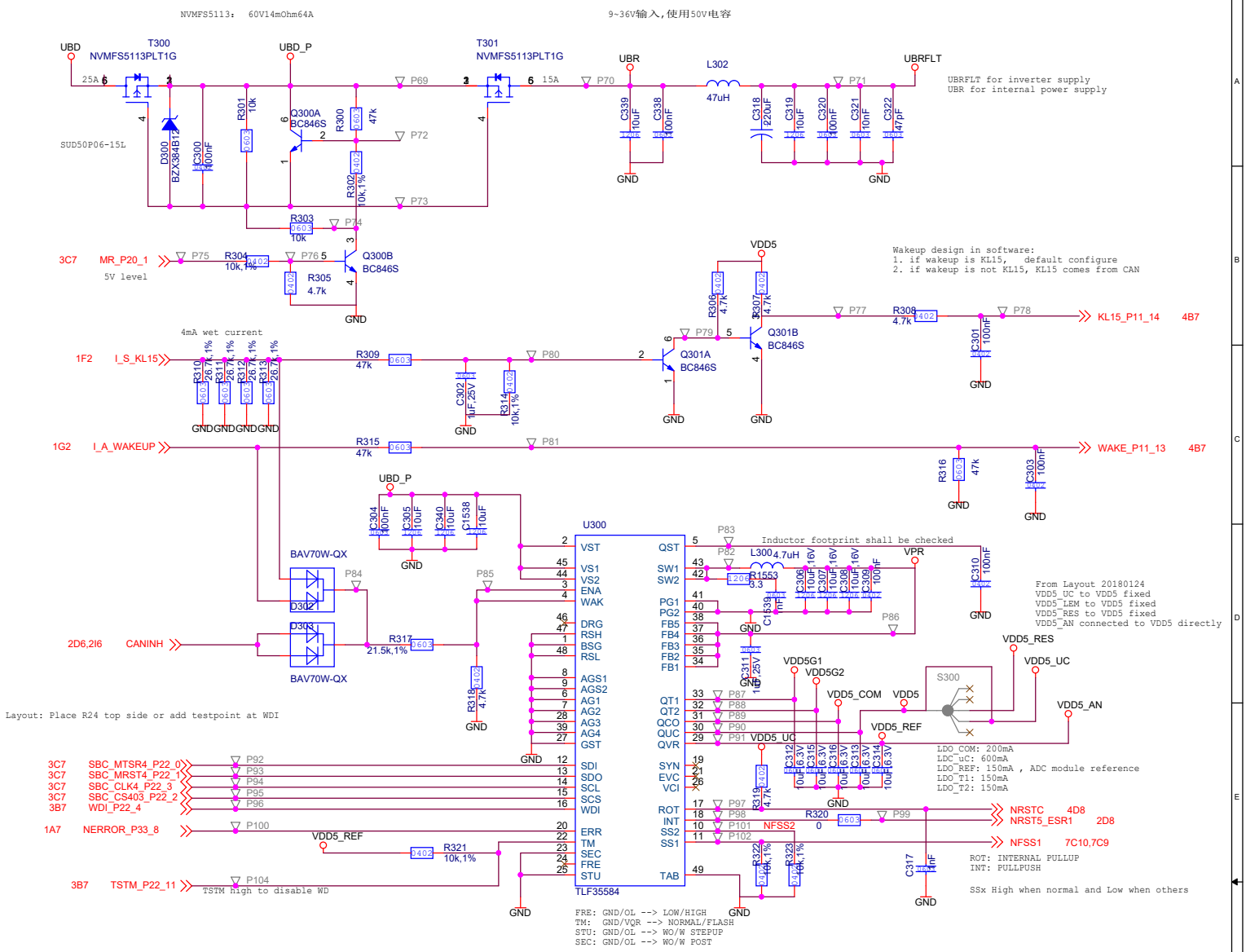
IMD

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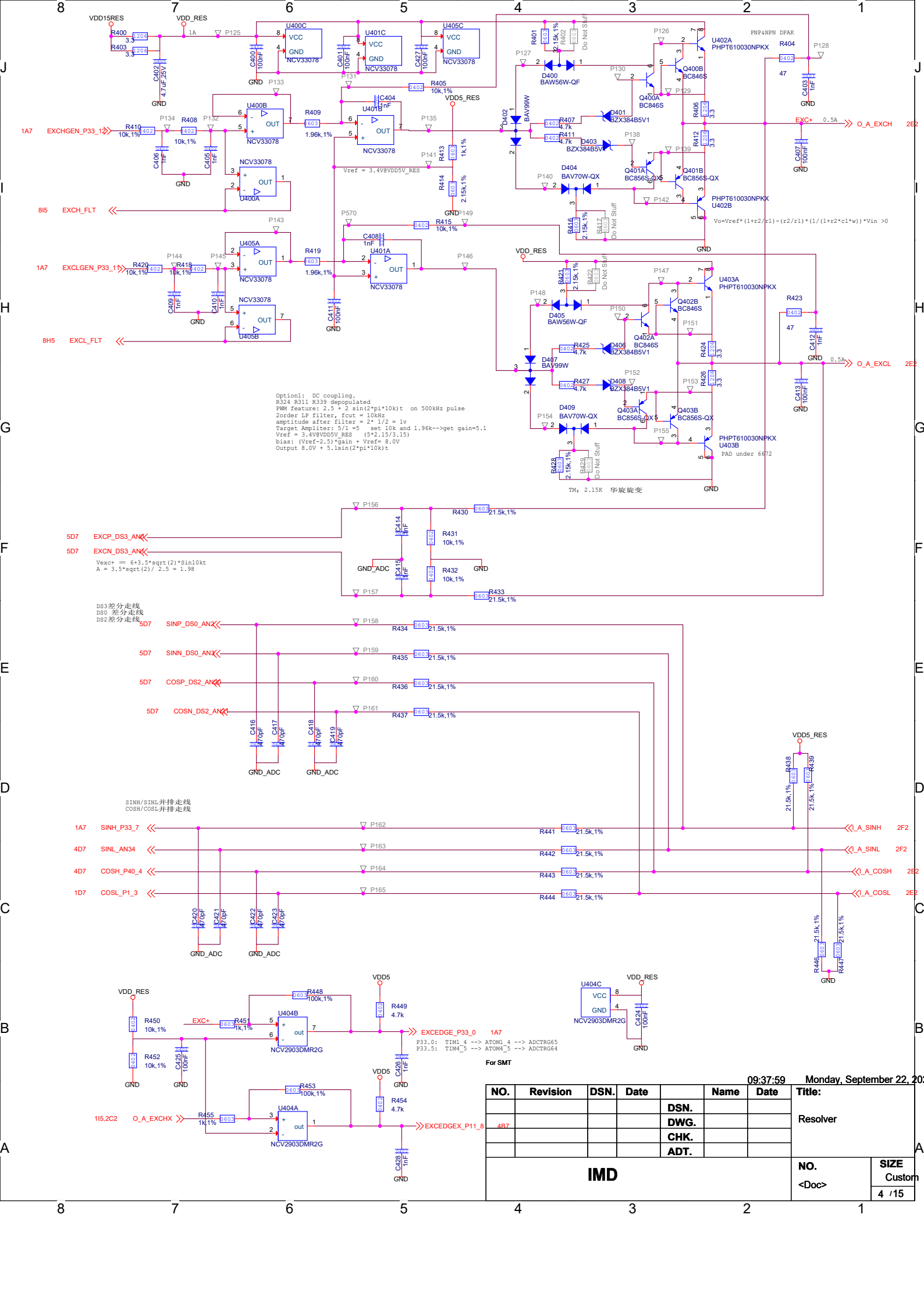
SIZE Custom

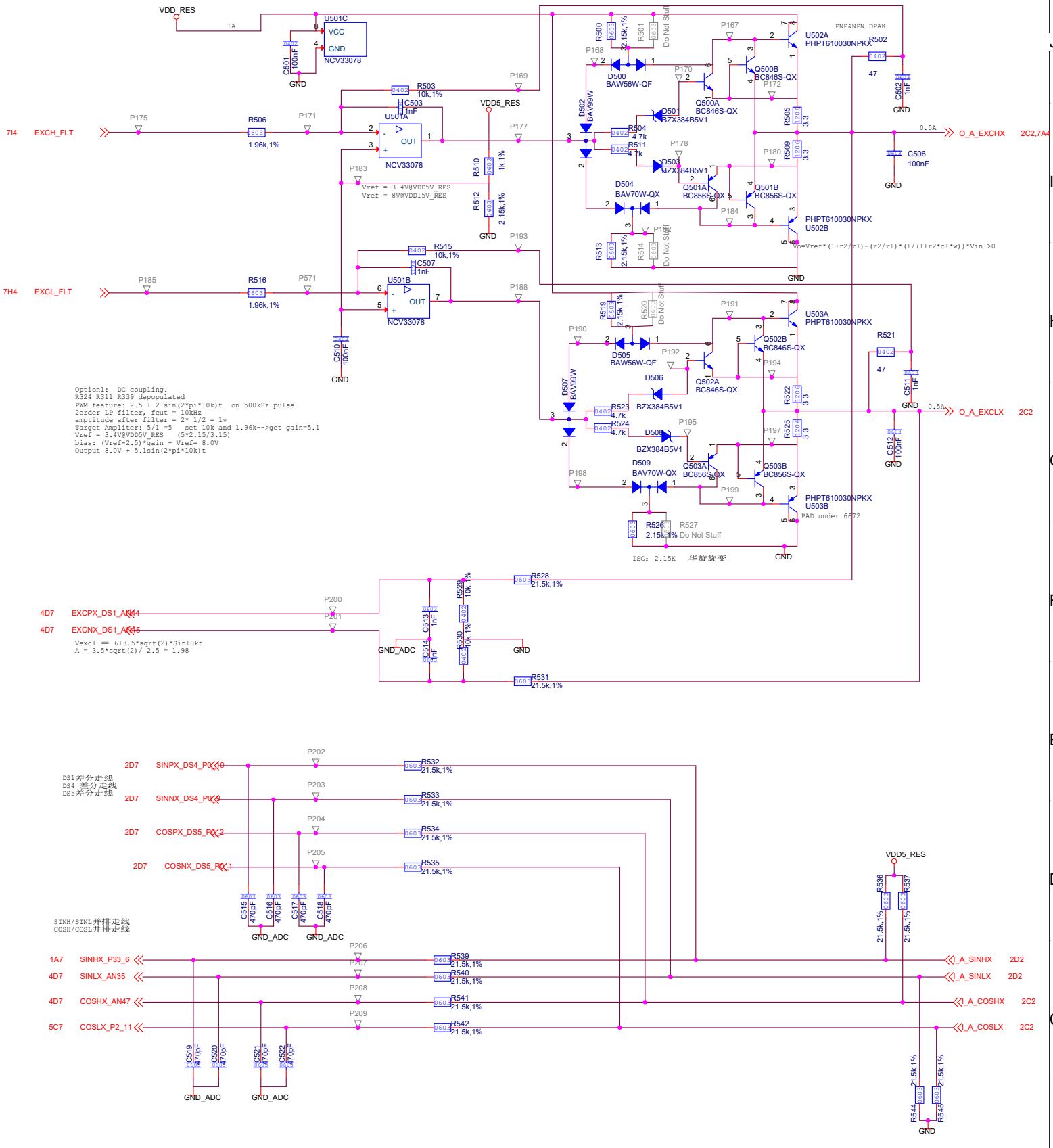
1 /15


X200 X201 X202 X203 X204 X205 X206 X207 X208
M4
GND
TRES1: Brake resistor temperature1
TRES1RT: Brake resistor temperature1 return
PS5V: Pressure sensor supply 5V
PS1 PS2: Pressure sensor1 2
PSRT: Pressure sensor supply return
J1503 忽略插件symbol中的名称
SMC20-P502B1-00
TM_EXCLO TM_SINHI TM_SINHI TM_COSHI ISG_COSHI TM_EMT0 ISG_EMT0RT TRES1 TRES2RT TM_SHLD ACANL ACANH CAN_SHLD CCANH CCANL HVIL- HVIL+ WAKEUP KL31 KL31 KL30 KL30
EXCL EXCL_X SINH SINH_X COSH COSH_X EMT0 EMT0RT_X TRES1 TRES2RT SHIELDAN ACANL ACANH SHIELDCAN VDD15 VDD15RT CCANH CCANL HVIL- HVIL+ WAKE KL31_4 KL31_3 KL30_4 KL30_3
EXCH EXCH_X SINL SINL_X COSL COSL_X EMT0RT EMT0_X TRES1RT TRES2 SHIELDAN_X PS1 PSRT PS2 PS5V SHIELDCAN2 CRASH BCANH BCANL BSL KL15 KL31 KL31_2 KL30 KL30_2
TM_EXCHI ISG_EXCHI TM_SINLO ISG_SINLO TM_COSLO ISG_COSLO TM_EMT0RT ISG_EMT0 TRES1RT TRES2 ISG_SHLD PS1 PSRT PS2 PS5V CAN_SHLD CRASH BCANH BCANL BSL KL15 KL31 KL31 KL30 KL30
VDD15FLUX
R204
BMSCANH
BMSCANL
C237
4.7uF,25V
删除了BMS接口， 保留一个供电可能
Schematic plan:
1) Control board: Page1~Page19
2) Drive board: Page20~Page29
3) Drive boardx: Page30~Page39
Pin1 close to board edge
L1206
15uH
KL30
P20
UBD
7A
D1524
TPSMB36CA
KL31
CAN_SHLD
TM_SHLD
ISG_SHLD
C1200 10uF
C1201 1uF
C1202 1nF
4.7uF 0805
10uF 1206
22uF 1210
Internal connectors to current sensors
Shuntsensor
5602200420 Male
5601230400 Female
Mate with 1473672-1
Pin1: GND
Pin2: CANH
Pin3: CANL
Pin4: VCC
ACANH ACANL BCANH BCANL CCANH CCANL
ACAN: Communication
BCAN: Calibration and internal
CCAN: Reserved in Con2
TVS deleted as the tranceiver ESD immunity
B_D_ACANH 8J6
B_D_ACANL 8J6
B_D_BCANH 8G6
B_D_BCANL 8G6
B_D_CCANH 8E6
B_D_CCANL 8E6
C203 C204 C205 C206 C207 C208
10nF
VDD5G2
Possible for pressure sensor
Reserved for inlet coolant temperature and oil temperature
PS5V PS1 PS2 PSRT
I_A_PRES1 7C12
I_A_TOIL 7C12
C1524 C1525 C209 C210
GND_ADC
0.5mm wire GND_ADC to connector from CPU
then 0.16mm to xxxRT pin opposite the connector side
BSL WAKEUP KL15
I_F_CFI 5C7
I_A_WAKEUP 2C3
I_S_KL15 2B3
C212 C213 C214
TM_SINLO TM_SINHI TM_COSHI TM_COSLO TM_EXCHI TM_EXCLO
R200 R201 R202 R203
ACT45B-510
L200 L201
Do Not Stuff
P61 P62 P63 P64
I_A_SINL 1C4
I_A_SINH 1D4
I_A_COSH 1C4
I_A_COSL 1C4
O_A_EXCH 1I4
O_A_EXCL 1G4
D200 D201 D202
NUP2105L
C215 C216 C217 C218 C219 C220
4.7nF/100V
1nF
C1523 100nF
ISG_SINLO ISG_SINHI ISG_COSHI ISG_COSLO ISG_EXCHI ISG_EXCLO
R1501 R1502 R1503 R1504
L1204 L1205
P65 P66 P67 P68
I_A_SINLX 1C5
I_A_SINHX 1C5
I_A_COSHX 1C5
I_A_COSLX 1C5
O_A_EXCHX 1I5,7A4
O_A_EXCLX 1G5
D1521 D1522 D1523
C1510 C1511 C1512 C1513 C1514 C1515
C1522 100nF
HVIL- HVIL+ TM_EMT0 TM_EMT0RT ISG_EMT0 ISG_EMT0RT TRES1 TRES1RT TRES2 TRES2RT
HVIL- 7B15
HVIL+ 7B15
I_A_EMT0 7H12
M_A_EMT0 7H12
I_A_EMT1 7G12
M_A_EMT1 7G12
I_A_TRES1 8J12
I_A_TRES2 8J12
Coolingplate
C228 C229 C230 C231 C232 C233 C234 C235 C236 C240 C241
GND_ADC
0.5mm wire GND_ADC to connector from CPU
then 0.16mm to xxxRT pin opposite the connector side
For SMT
09:37:59
Monday, September 22,
NO. Revision DSN. Date Name Date
DSN.
DWG.
CHK.
ADT.
Title:
External Interface
IMD
NO.
<Doc>
SIZE
Custor
2 /15



For SMT							09:37:59	Monday, September 22, 2025
NO.	Revision	DSN.	Date		Name	Date	Title:	
				DSN.				
				DWG.			Power Supply	
				CHK.				
				ADT.				
IMD							NO. <Doc>	SIZE Custom
								3 /15


VDD15RES
VDD_RES
U400C
U401C
U405C
NCV33078
U400B
U401B
U400A
U405A
U405B
EXCHGEN_P33_12
EXCLGEN_P33_1
EXCH_FLT
EXCL_FLT
VDD5_RES
Vref = 3.4V@VDD5V_RES
Option1: DC coupling.
R324 R311 R339 depopulated
PWM feature: 2.5 + 2 sin(2*pi*10k)t on 500kHz pulse
2order LP filter, fcut = 10kHz
amptitude after filter = 2* 1/2 = 1v
Target Ampliter: 5/1 =5 set 10k and 1.96k-->get gain=5.1
Vref = 3.4V@VDD5V_RES (5*2.15/3.15)
bias: (Vref-2.5)*gain + Vref= 8.0V
Output 8.0V + 5.1sin(2*pi*10k)t
PNP&NPN DPAK
U402A PHPT610030NPKX
U400B BC846S
Q400A BC846S
D400 BAW56W-QF
D401 BZX384B5V1
D403 BZX384B5V1
D404 BAV70W-QX
Q401A BC856S-QX
Q401B BC856S-QX
U402B PHPT610030NPKX
Vo=Vref*(1+r2/r1)-(r2/r1)*(1/(1+r2*cl*w))*Vin >0
U403A PHPT610030NPKX
Q402B BC846S
Q402A BC846S
D405 BAW56W-QF
D406 BZX384B5V1
D407 BAV99W
D408 BZX384B5V1
D409 BAV70W-QX
Q403A BC856S-QX
Q403B BC856S-QX
U403B PHPT610030NPKX
PAD under 6672
TM: 2.15K 华旋旋变
O_A_EXCH
O_A_EXCL
EXC+ 0.5A
Vexc+ = 6+3.5*sqrt(2)*Sin10kt
A = 3.5*sqrt(2)/ 2.5 = 1.98
EXCP_DS3_AN4
EXCN_DS3_AN4
DS3差分走线
DS0 差分走线
DS2差分走线
SINP_DS0_AN2
SINN_DS0_AN3
COSP_DS2_AN0
COSN_DS2_AN1
SINH/SINL并排走线
COSH/COSL并排走线
SINH_P33_7
SINL_AN34
COSH_P40_4
COSL_P1_3
I_A_SINH
I_A_SINL
I_A_COSH
I_A_COSL
U404B
U404A
U404C
NCV2903DMR2G
EXCEDGE_P33_0
EXCEDGEX_P11_8
P33.0: TIM1_4 --> ATOM1_4 --> ADCTRG65
P33.5: TIM4_5 --> ATOM4_5 --> ADCTRG64
For SMT
09:37:59 Monday, September 22, 202
NO. Revision DSN. Date Name Date
DSN.
DWG.
CHK.
ADT.
Title:
Resolver
IMD
NO.
<Doc>
SIZE
Custom
4 /15



For SMT

09:41:58 Monday, September 22, 202

NO.	Revision	DSN.	Date		Name	Date
				DSN.		
				DWG.		
				CHK.		
				ADT.		

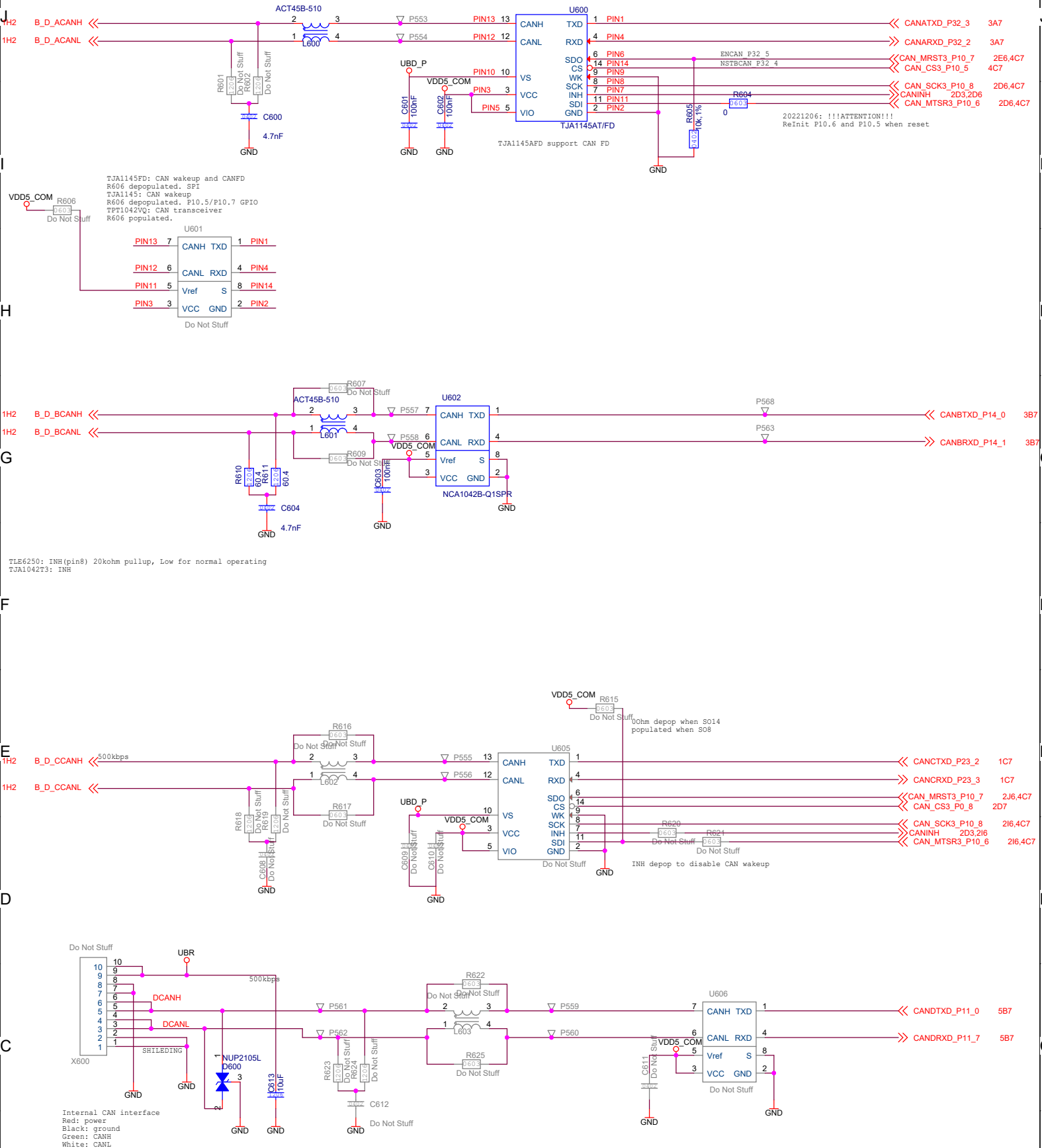
Title: 2nd Resolver

IMD

NO. <Doc>

SIZE Custom

5 /15



DCDC: 0430450400 0430250400
1-1: CANL
1-2: ENABLE
1-3: CANH
1-4: 12V/24V
2-1: GND
2-2: PAN+ 12V

ISO: (1-1827876-4)
1. CANH
2. CANGND
3. MSD-
4. VIN
5. CANL
6. MSD+
7. KE
8. GND

For SMT

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NO.	Revision	DSN.	Date		Name	Date
				DSN.		
				DWG.		
				CHK.		
				ADT.		

Title: Communication

IMD

NO. <Doc>

SIZE Custom

6 /15

OPL 20181004:
DRV SPI changed to P11, need update HSI --> Change other port if possible
ENH/ENL shall be removed from P10.5 P10.6
SBC SPI occupy the P22.0~3, GPIO will reassigned

B0 sample
1) ENDRVH从P32_7换到P32_5
2) PWMTRGX从P14_5换到P32_7
3) add adctest circuit on AN18
4) TEMX moved to AN29 from AN18

U700F
Analog Inputs
ADC Groups
Group 0
Group 1
Group 2
Group 3/8 Sent/Port 40
Group 3/4/8 Sent/Port 40
Group 4/8
TC387

U700G
Port 0 & 1
GPIO/GTM/QSPI3/CAN
GTM/CCU6/ASC/CAN
4X 100kHz
TC387

U700K
Port 20, 21 & 15
GTM/ASC1/QSPI0,2/CAN4
CCU6/GPT120/I2C0
GTM/ASC3/LVDS/HSCT/QSPI2,3
QSPI0 in P20_8~14
TC387

U700H
Port 2 & 10
GPIO/QSPI/CCU6/VADCEMUX
GPT/GTM/QSPI1/CCU6
TC387

U700L
Port 22 & 23
GTM/ASC3/LVDS/HSCT/QSPI2,3
TC387

U700I
Port 11
GTM/ASC3/CCU6/GPT120
TC387

U700J
Port 12, 13 & 14
GTM/I2C0/QSPI2/CAN/CCU6/GPT120
TC387

U700M
Port 32, 33 & 34
ASC2/GTM/SENT/DSADC/VADCMUX/CCU6
TC387
Port33/34 is VEVRSB

P13_0 : MSC_CLKN
P13_1 : MSC_CLKP
P13_2 : MSC_TXDN
P13_3 : MSC_TXDP

P13.0 TOM0_5 TOUT91
P13.1 TOM0_6 TOUT92
P13.2 TOM0_7 TOUT93
P13.3 TOM0_8 TOUT94

CFG0: DN
CFG1: UP
CFG2: UP when 3Cores. DN when 4+cores
CFG3: DN
CFG4: DN
CFG5: DEFAULT, CAN BE DN
CFG6: UP

HBxy: 10kHz
SINx: 10kHz
COSx: 10kHz
EXCx: 10kHz
PWMxy: 20kHz
PPUP: 100kHz
PPDN: 100kHz
CAN: 1MHz
EEP SPI: 1MHz
LSD SPI: 2MHz
TLF SPI: 5MHz
FFxSPI: 500kHz
f>=500kHz shall be shielded

OutRing
Column1/2/29/30
RowA/B/AJ/AK

HWCFG4:5
01, ABM WHEN P14.X FAIL,
TARGET BOOT CONFIGURE

HWCFG[0] P14.6	HWCFG[1] P14.5	HWCFG[2] P14.2	HWCFG[3] P14.3	HWCFG[4] P10.5	HWCFG[5] P10.6	HWCFG[6] P14.4
0 - SMPS 1 - LDO (default)	0 - EVR33OFF 1 - EVR33ON (default)	0 - EVR13OFF 1 - EVR13ON (default)	0 - Boot from pins 1 - Flash BMI boot (default)	HWCFG[4:5] [0 0]- Generic Bootstrap (P14.0/1) [0 1]- ABM, Generic Bootstrap on fail (P14.0/1) [1 0]- ABM, ASC Bootstrap on fail (P15.2/3) [1 1]- Internal start from Flash (default)		Default Pad state 0 - Pins in tristate 1 - Pins with pull-up (default)

For SMT

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NO.	Revision	DSN.	Date		Name	Date
				DSN.		
				DWG.		
				CHK.		
				ADT.		

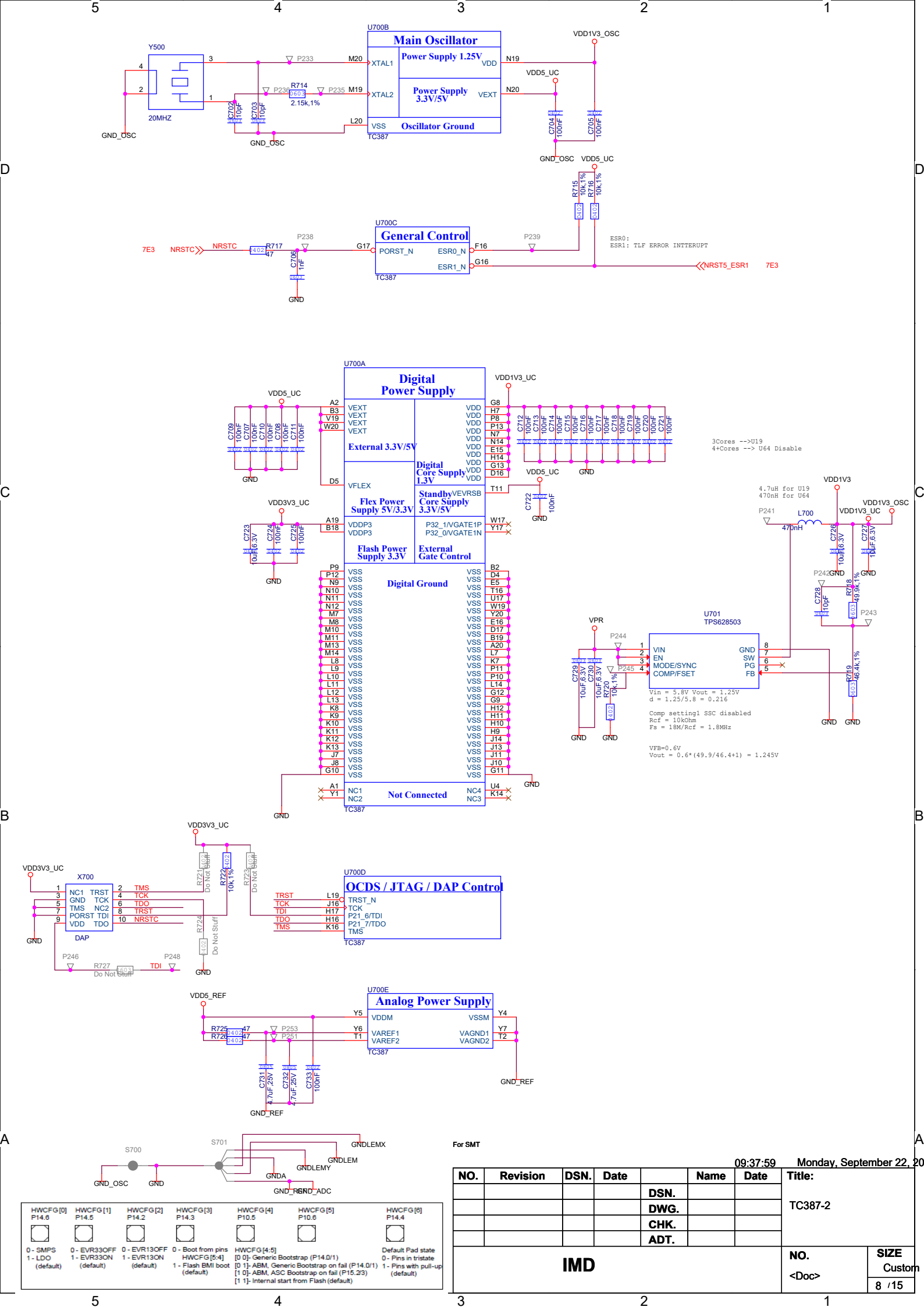
Title: TC387-1

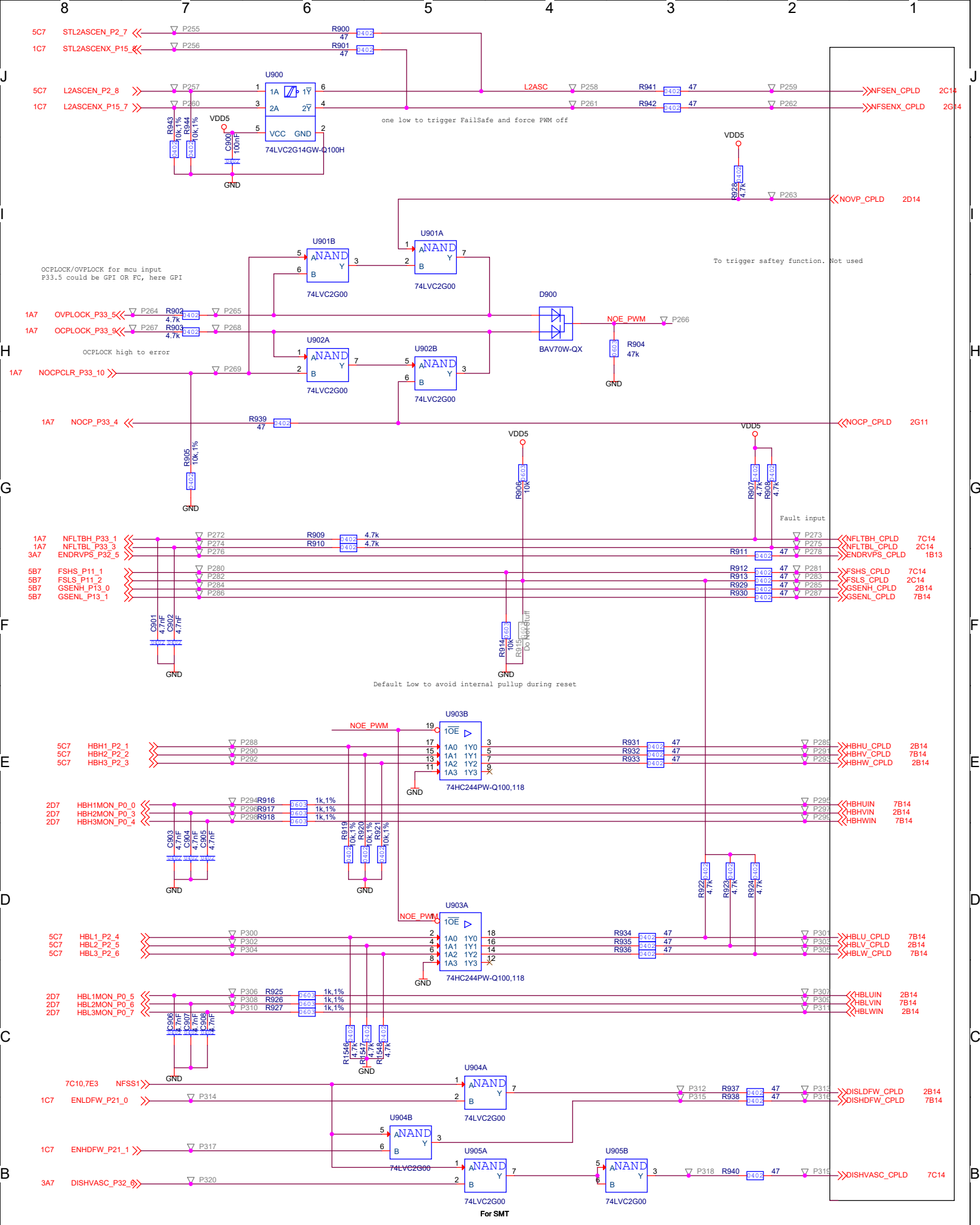
IMD

NO. <Doc>

SIZE Custom

7 /15


U700B
Main Oscillator
Power Supply 1.25V
Power Supply 3.3V/5V
Oscillator Ground
TC387
Y500
20MHZ
R714
2.15k,1%
C702 10pF
C703 10pF
GND_OSC
VDD1V3_OSC
VDD5_UC
C704 100nF
C705 100nF
U700C
General Control
PORST_N
ESR0_N
ESR1_N
R715 10k,1%
R716 10k,1%
ESR0:
ESR1: TLF ERROR INTTERUPT
NRSTC
R717 47
C706 1nF
NRST5_ESR1
U700A
Digital Power Supply
External 3.3V/5V
Digital Core Supply 1.3V
Flex Power Supply 5V/3.3V
Standby Core Supply 3.3V/5V
Flash Power Supply 3.3V
External Gate Control
Digital Ground
Not Connected
C709 100nF
C707 100nF
C710 100nF
C708 100nF
C711 100nF
C712 100nF
C713 100nF
C714 100nF
C715 100nF
C716 100nF
C717 100nF
C718 100nF
C719 100nF
C720 100nF
C721 100nF
C722 100nF
C723 10uF/6.3V
C724 100nF
C725 100nF
3Cores -->U19
4+Cores --> U64 Disable
4.7uH for U19
470nH for U64
L700 470nH
C726 10uF/6.3V
C727 10uF/6.3V
C728 10pF
R718 49.9K,1%
R719 46.4K,1%
U701
TPS628503
VIN
EN
MODE/SYNC
COMP/FSET
GND
SW
PG
FB
C729 10uF/6.3V
C730 10uF/6.3V
R720 10k,1%
Vin = 5.8V Vout = 1.25V
d = 1.25/5.8 = 0.216
Comp setting1 SSC disabled
Rcf = 10kOhm
Fs = 18M/Rcf = 1.8MHz
VFB=0.6V
Vout = 0.6*(49.9/46.4+1) = 1.245V
X700
DAP
TMS
TCK
TDO
TRST
NRSTC
TDI
R721 Do Not Stuff
R722 10k,1%
R723 Do Not Stuff
R724 Do Not Stuff
R727 Do Not Stuff
U700D
OCDS / JTAG / DAP Control
TRST_N
P21_6/TDI
P21_7/TDO
U700E
Analog Power Supply
VDDM
VSSM
VAREF1
VAREF2
VAGND1
VAGND2
R725 47
R726 47
C731 4.7uF/25V
C732 4.7uF/25V
C733 100nF
GND_REF
S700
S701
GNDLEMX
GNDLEM
GNDLEMY
GNDA
GND_ADC
HWCFG[0] P14.6
0 - SMPS
1 - LDO (default)
HWCFG[1] P14.5
0 - EVR33OFF
1 - EVR33ON (default)
HWCFG[2] P14.2
0 - EVR13OFF
1 - EVR13ON (default)
HWCFG[3] P14.3
0 - Boot from pins
1 - Flash BMI boot (default)
HWCFG[4] P10.5
HWCFG[5] P10.6
HWCFG[4:5]
[0 0]- Generic Bootstrap (P14.0/1)
[0 1]- ABM, Generic Bootstrap on fail (P14.0/1)
[1 0]- ABM, ASC Bootstrap on fail (P15.2/3)
[1 1]- Internal start from Flash (default)
HWCFG[6] P14.4
Default Pad state
0 - Pins in tristate
1 - Pins with pull-up (default)
For SMT
09:37:59 Monday, September 22, 20
NO. Revision DSN. Date Name Date
DSN.
DWG.
CHK.
ADT.
Title:
TC387-2
IMD
NO.
<Doc>
SIZE
Custom
8 /15



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NO.	Revision	DSN.	Date		Name	Date
				DSN.		
				DWG.		
				CHK.		
				ADT.		

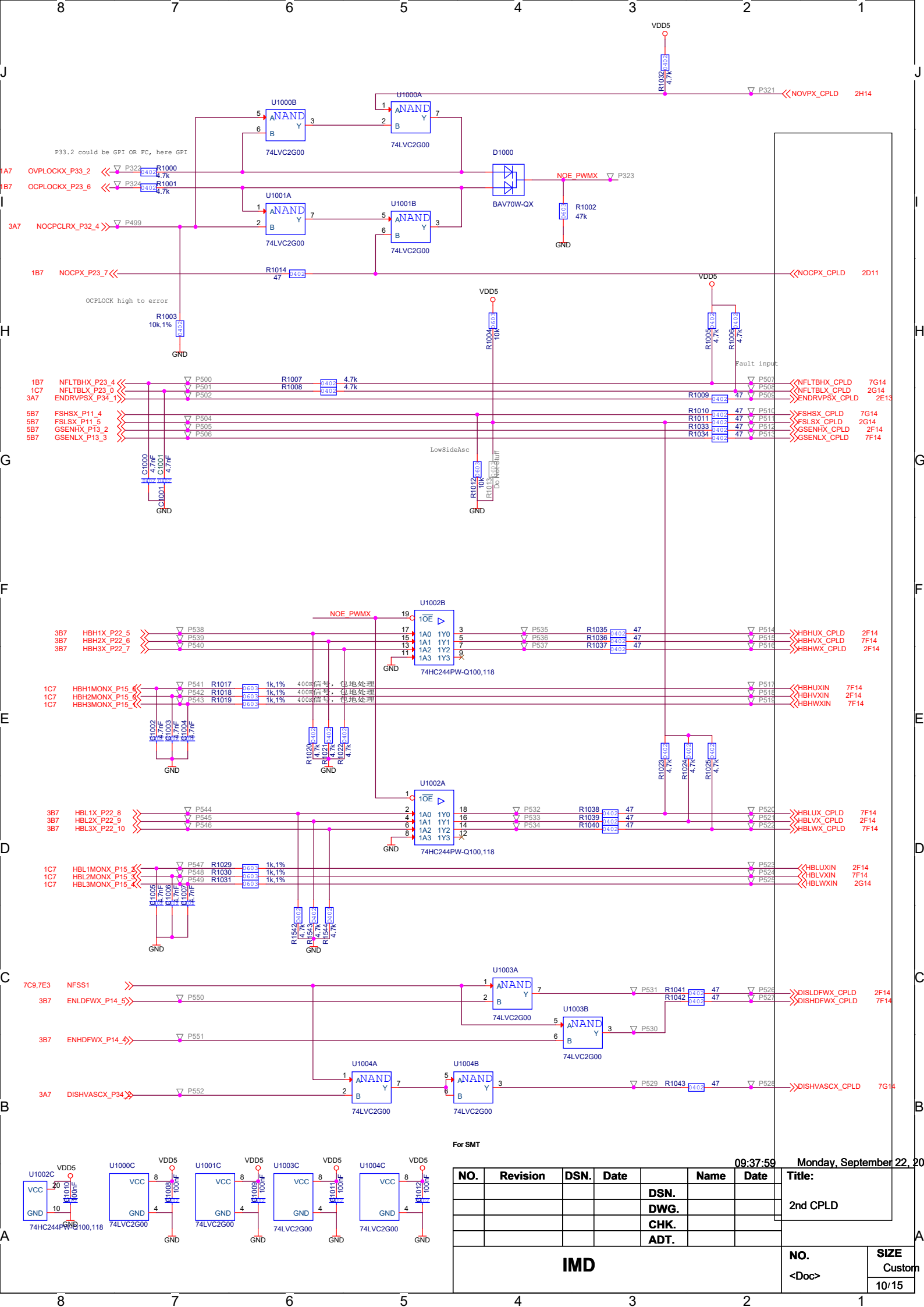
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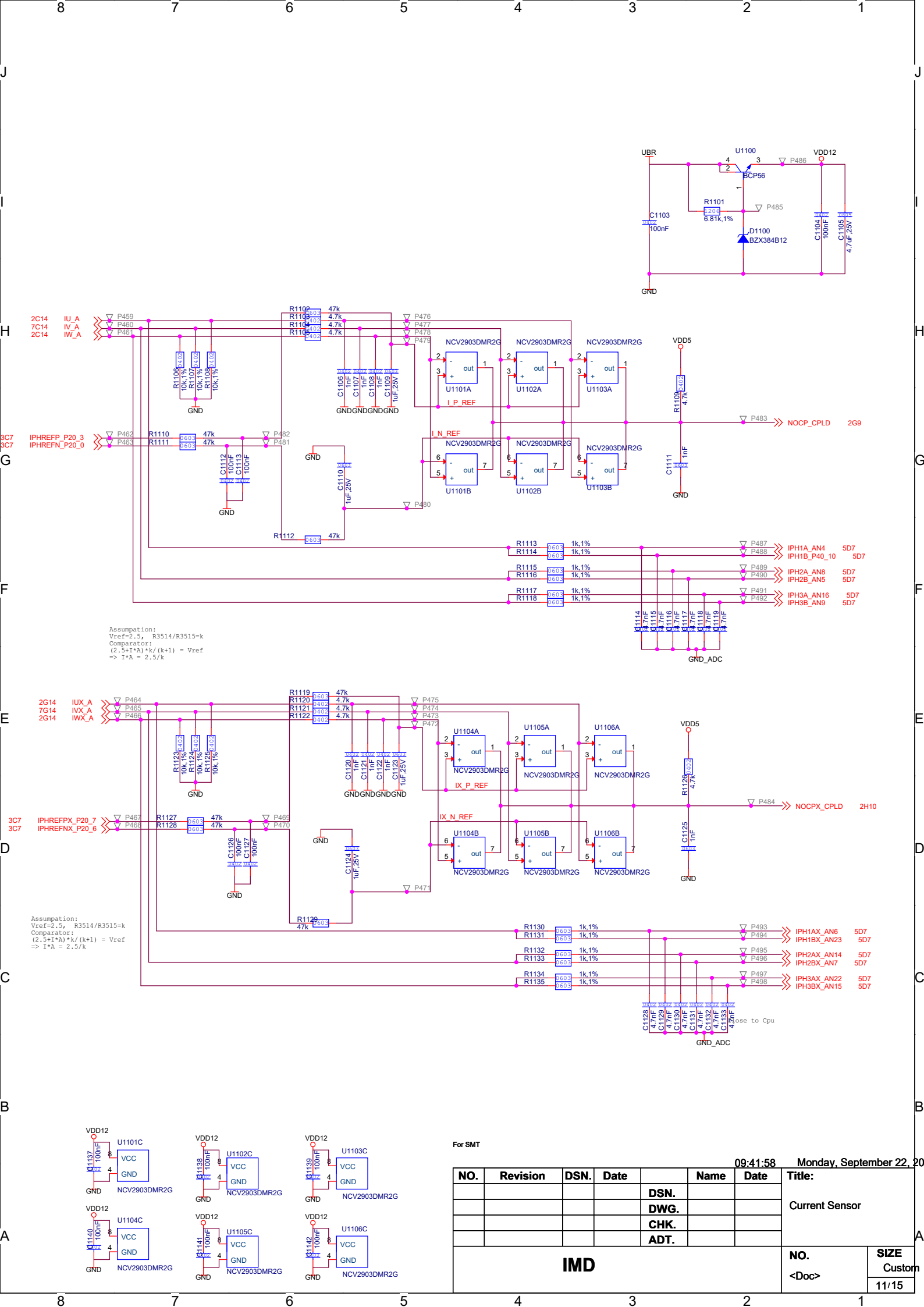
IMD

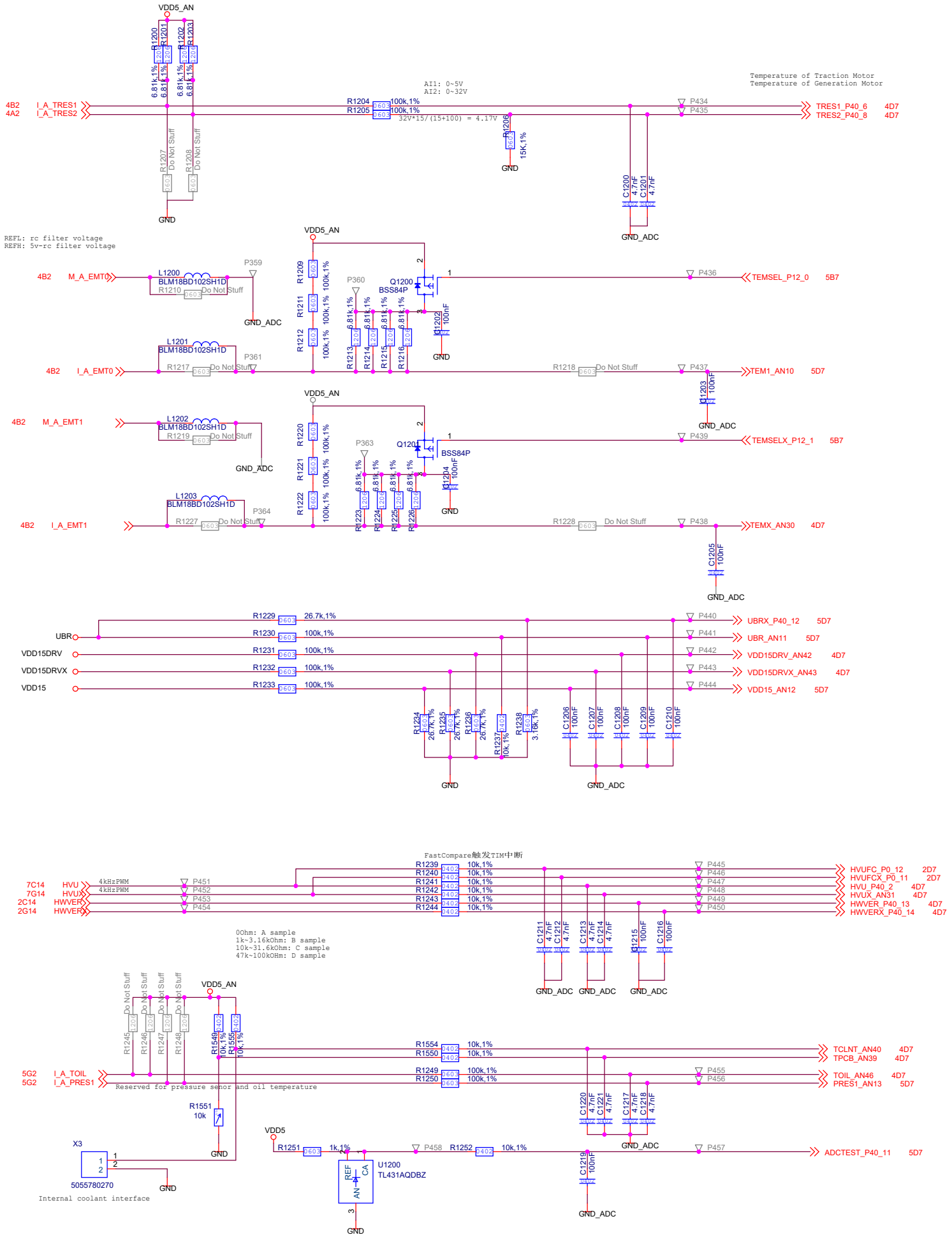
NO. <Doc>

SIZE Custom

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VDD5
R1032 4.7K
P321
NOVPX_CPLD 2H14
U1000A
U1000B
74LVC2G00
P33.2 could be GPI OR FC, here GPI
OVPLOCKX_P33_2
OCPLOCKX_P23_6
R1000 4.7k
R1001 4.7k
D1000
BAV70W-QX
NOE_PWMX
P323
R1002 47k
U1001A
U1001B
NOCPCLRX_P32_4
P499
NOCPX_P23_7
R1014 47
NOCPX_CPLD 2D11
OCPLOCK high to error
R1003 10k,1%
GND
R1004 10k
R1005 4.7k
R1006 4.7k
Fault input
NFLTBHX_P23_4
NFLTBLX_P23_0
ENDRVPSX_P34_1
R1007 4.7k
R1008 4.7k
R1009 47
NFLTBHX_CPLD 7G14
NFLTBLX_CPLD 2G14
ENDRVPSX_CPLD 2E13
FSHSX_P11_4
FSLSX_P11_5
GSENHX_P13_2
GSENLX_P13_3
R1010 47
R1011 47
R1033 47
R1034 47
FSHSX_CPLD 7G14
FSLSX_CPLD 2G14
GSENHX_CPLD 2F14
GSENLX_CPLD 7F14
C1000 4.7nF
C1001 4.7nF
LowSideAsc
R1012 10k
R1013 Do Not Stuff
U1002B
U1002A
74HC244PW-Q100,118
HBH1X_P22_5
HBH2X_P22_6
HBH3X_P22_7
R1035 47
R1036 47
R1037 47
HBHUX_CPLD 2F14
HBHVX_CPLD 7F14
HBHWX_CPLD 2F14
HBH1MONX_P15_6
HBH2MONX_P15_6
HBH3MONX_P15_6
R1017
R1018
R1019
1k,1%
400K信号，包地处理
C1002 4.7nF
C1003 4.7nF
C1004 4.7nF
R1020 4.7K
R1021 4.7K
R1022 4.7K
R1023 4.7K
R1024 4.7K
R1025 4.7K
HBHUXIN 7F14
HBHVXIN 2F14
HBHWXIN 7F14
HBL1X_P22_8
HBL2X_P22_9
HBL3X_P22_10
R1038 47
R1039 47
R1040 47
HBLUX_CPLD 7F14
HBLVX_CPLD 2F14
HBLWX_CPLD 7F14
HBL1MONX_P15_4
HBL2MONX_P15_4
HBL3MONX_P15_4
R1029
R1030
R1031
C1005 4.7nF
C1006 4.7nF
C1007 4.7nF
R1542 4.7K
R1543 4.7K
R1544 4.7K
HBLUXIN 2F14
HBLVXIN 7F14
HBLWXIN 2G14
NFSS1
ENLDFWX_P14_5
ENHDFWX_P14_4
DISHVASCX_P34
U1003A
U1003B
U1004A
U1004B
R1041 47
R1042 47
R1043 47
DISLDFWX_CPLD 2F14
DISHDFWX_CPLD 7F14
DISHVASCX_CPLD 7G14
U1002C
U1000C
U1001C
U1003C
U1004C
VCC
GND
C1010 100nF
C1008 100nF
C1009 100nF
C1011 100nF
C1012 100nF
For SMT
09:37:59 Monday, September 22, 20
NO. Revision DSN. Date Name Date
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DWG.
CHK.
ADT.
Title:
2nd CPLD
IMD
NO.
<Doc>
SIZE
Custom
10/15


UBR
U1100
BCP56
VDD12
P486
R1101
6.81k,1%
P485
C1103
100nF
D1100
BZX384B12
C1104
100nF
C1105
4.7uF,25V
GND
2C14 IU_A
7C14 IV_A
2C14 IW_A
P459
P460
P461
R1102 47k
R1103 4.7k
R1104 4.7k
R1105 4.7k
P476
P477
P478
P479
R1106 10k,1%
R1107 10k,1%
R1108 10k,1%
C1106 1nF
C1107 1nF
C1108 1nF
C1109 1uF,25V
NCV2903DMR2G
U1101A
U1102A
U1103A
VDD5
R1109 4.7k
I_P_REF
I_N_REF
P483
NOCP_CPLD 2G9
3C7 IPHREFP_P20_3
3C7 IPHREFN_P20_0
P462
P463
R1110 47k
R1111 47k
P482
P481
C1112 100nF
C1113 100nF
C1110 1uF,25V
U1101B
U1102B
U1103B
C1111 1nF
P480
R1112 47k
R1113 1k,1%
R1114 1k,1%
R1115 1k,1%
R1116 1k,1%
R1117 1k,1%
R1118 1k,1%
P487
P488
P489
P490
P491
P492
IPH1A_AN4 5D7
IPH1B_P40_10 5D7
IPH2A_AN8 5D7
IPH2B_AN5 5D7
IPH3A_AN16 5D7
IPH3B_AN9 5D7
C1114 4.7nF
C1115 4.7nF
C1116 4.7nF
C1117 4.7nF
C1118 4.7nF
C1119 4.7nF
GND_ADC
Assumpation:
Vref=2.5, R3514/R3515=k
Comparator:
(2.5+I*A)*k/(k+1) = Vref
=> I*A = 2.5/k
2G14 IUX_A
7G14 IVX_A
2G14 IWX_A
P464
P465
P466
R1119 47k
R1120 4.7k
R1121 4.7k
R1122 4.7k
P475
P474
P473
P472
R1123 10k,1%
R1124 10k,1%
R1125 10k,1%
C1120 1nF
C1121 1nF
C1122 1nF
C1123 1uF,25V
U1104A
U1105A
U1106A
IX_P_REF
IX_N_REF
P484
NOCPX_CPLD 2H10
3C7 IPHREFPX_P20_7
3C7 IPHREFNX_P20_6
P467
P468
R1127 47k
R1128 47k
P469
P470
C1126 100nF
C1127 100nF
C1124 1uF,25V
U1104B
U1105B
U1106B
P471
R1129 47k
R1130 1k,1%
R1131 1k,1%
R1132 1k,1%
R1133 1k,1%
R1134 1k,1%
R1135 1k,1%
P493
P494
P495
P496
P497
P498
IPH1AX_AN6 5D7
IPH1BX_AN23 5D7
IPH2AX_AN14 5D7
IPH2BX_AN7 5D7
IPH3AX_AN22 5D7
IPH3BX_AN15 5D7
C1128 4.7nF
C1129 4.7nF
C1130 4.7nF
C1131 4.7nF
C1132 4.7nF
C1133 4.7nF
Close to Cpu
GND_ADC
Assumpation:
Vref=2.5, R3514/R3515=k
Comparator:
(2.5+I*A)*k/(k+1) = Vref
=> I*A = 2.5/k
U1101C
U1102C
U1103C
U1104C
U1105C
U1106C
VCC
GND
NCV2903DMR2G
C1137 100nF
C1138 100nF
C1139 100nF
C1140 100nF
C1141 100nF
C1142 100nF
For SMT
09:41:58 Monday, September 22, 20
NO.
Revision
DSN.
Date
Name
Date
DSN.
DWG.
CHK.
ADT.
Title:
Current Sensor
IMD
NO.
<Doc>
SIZE
Custom
11/15



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NO.	Revision	DSN.	Date		Name	Date
				DSN.		
				DWG.		
				CHK.		
				ADT.		

Title: Analog

IMD

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Truth table

Set	Reset	Clock	Data	Q
L	H	X	X	H
H	H	UP	L	L
H	H	H	X	No change
H	H	L	X	No change

1) PMOS Default on:
SET pull down by resistor (truth table 1)
2) ENDRVPS = H. PMOS no change, keep on
Clock = L as Vref<VDD15DRV(truth table 4)
3) Vref>VDD15DRV, PMOS off:
Clock rising edge (truth table 2)
4) ENDRVPS = L. PMOS on:
SET pull down by gpio(truth table 1)
5) ENDRVPS = H. PMOS no change, keep on
SET pull up by gpio(truth table 4)

0Ohm to bypass efuse function
Depopulated

VDD15 — R1206 R1300 0,1% — VDD15DRV / VDD15HS / VDD15LS
Q1300 Do Not Stuff
R1301 10k,1% — R1302 4.7k
P378 P379 P380

2F9 ENDRVPS_CPLD

Low to bypass eFuse
High to enable eFuse, and OFF when OCP

P381 — R1304 4.7k — P382 — Q1301A BC846S — R1305 10k,1% — GND

D1525 BAV70W-QX
Close to MOSFET
VDD15 — P385 — R1307 10k,1% — P384 — R1308 10k,1% — C1300 100nF — GND
Vref = (VDD15-Vf)/2 (Vf=0.7V)

VDD15DRV — R1309 10k,1% — U1301B NCV2903DMR2G — R1306 4.7k — VDD5
R1310 10k,1% — C1305 100nF — GND
C1301 4.7nF — GND
P383 — R1311 4.7k — GND

VDD5 — U1300B 74HC74PW-Q100 (RST2, D2, CLK2, SET2, Q2, Q2)

VDD15=(VDD15 - Rdson * I)/2
When VDD15HS<Vref, a rising edge at CLK
Threshold:
Ith = Vf/Rdson

0Ohm to bypass efuse function
Depopulated

VDD15 — R1206 R1312 0,1% — VDD15DRVX / VDD15HSX / VDD15LSX
Q1302 Do Not Stuff
R1313 10k,1% — R1314 4.7k
P387 P388 P389

2G10 ENDRVPSX_CPLD

Low to bypass eFuse
High to enable eFuse, and OFF when OCP

P569 — R1316 4.7k — P391 — Q1301B BC846S — R1317 10k,1% — GND

VDD15DRVX — R1319 10k,1% — U1301A NCV2903DMR2G — R1318 4.7k — VDD5
R1320 10k,1% — C1306 100nF — GND
C1302 4.7nF — GND
P393 P392 — R1321 4.7k — GND

VDD5 — U1300A 74HC74PW-Q100 (RST1, D1, CLK1, SET1, Q1, Q1)

U1301C NCV2903DMR2G (VCC, GND) — VDD15
C1303 100nF — GND

U1300C 74HC74PW-Q100 (VCC, GND) — VDD5
C1304 100nF — GND

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				DSN.		
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				ADT.		

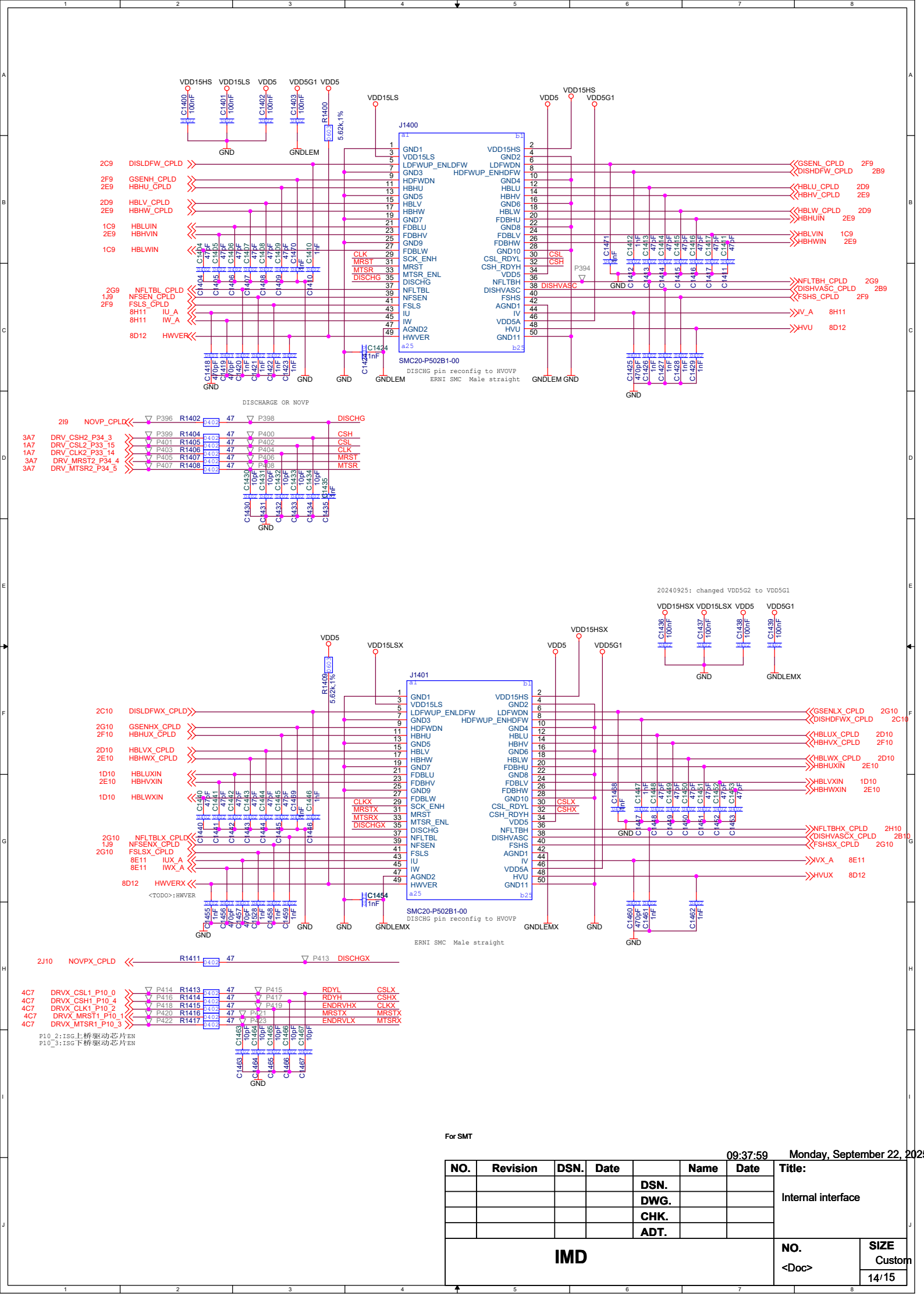
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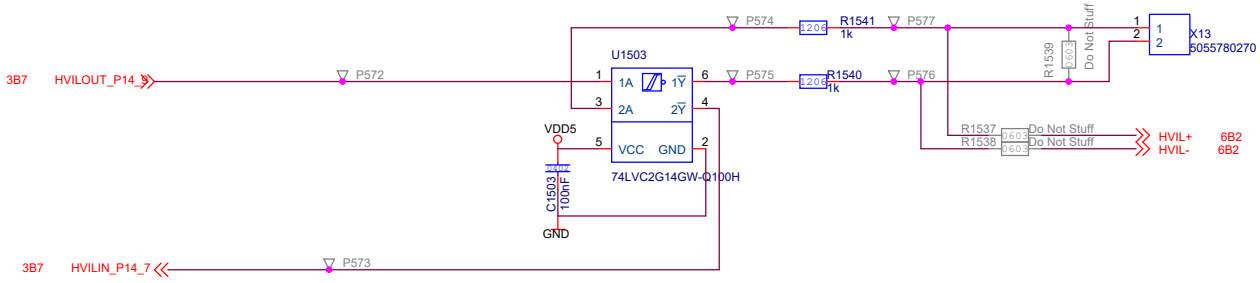
IMD

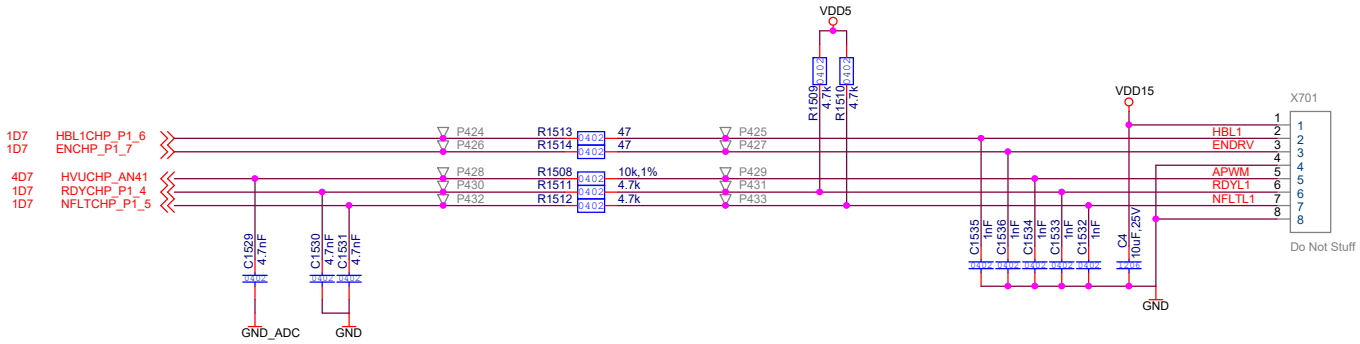
NO. <Doc>

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VDD15HS VDD15LS VDD5 VDD5G1 VDD5
C1400 100nF
C1401 100nF
C1402 100nF
C1403 100nF
R1400 5.62k,1%
GND
GNDLEM
VDD15LS
VDD5
VDD15HS
VDD5G1
J1400
GND1
VDD15LS
LDFWUP_ENLDFW
GND3
HDFWDN
HBHU
GND5
HBLV
HBHW
GND7
FDBLU
FDBHV
GND9
FDBLW
SCK_ENH
MRST
MTSR_ENL
DISCHG
NFLTBL
NFSEN
FSLS
IU
IW
AGND2
HWVER
VDD15HS
GND2
LDFWDN
HDFWUP_ENHDFW
GND4
HBLU
HBHV
GND6
HBLW
FDBHU
GND8
FDBLV
FDBHW
GND10
CSL_RDYL
CSH_RDYH
VDD5
NFLTBH
DISHVASC
FSHS
AGND1
IV
VDD5A
HVU
GND11
SMC20-P502B1-00
DISCHG pin reconfig to HVOVP
ERNI SMC Male straight
2C9 DISLDFW_CPLD
2F9 GSENH_CPLD
2E9 HBHU_CPLD
2D9 HBLV_CPLD
2E9 HBHW_CPLD
1C9 HBLUIN
2E9 HBHVIN
1C9 HBLWIN
2G9 NFLTBL_CPLD
1J9 NFSEN_CPLD
2F9 FSLS_CPLD
8H11 IU_A
8H11 IW_A
8D12 HWVER
CLK
MRST
MTSR
DISCHG
CSL
CSH
DISHVASC
P394
GSENL_CPLD 2F9
DISHDFW_CPLD 2B9
HBLU_CPLD 2D9
HBHV_CPLD 2E9
HBLW_CPLD 2D9
HBHUIN 2E9
HBLVIN 1C9
HBHWIN 2E9
NFLTBH_CPLD 2G9
DISHVASC_CPLD 2B9
FSHS_CPLD 2F9
IV_A 8H11
HVU 8D12
DISCHARGE OR NOVP
2I9 NOVP_CPLD
3A7 DRV_CSH2_P34_3
1A7 DRV_CSL2_P33_15
1A7 DRV_CLK2_P33_14
3A7 DRV_MRST2_P34_4
3A7 DRV_MTSR2_P34_5
R1402 47 DISCHG
R1404 47 CSH
R1405 47 CSL
R1406 47 CLK
R1407 47 MRST
R1408 47 MTSR
20240925: changed VDD5G2 to VDD5G1
VDD15HSX VDD15LSX VDD5 VDD5G1
C1436 100nF
C1437 100nF
C1438 100nF
C1439 100nF
GNDLEMX
VDD15LSX
VDD15HSX
J1401
CLKX
MRSTX
MTSRX
DISCHGX
CSLX
CSHX
2C10 DISLDFWX_CPLD
2G10 GSENHX_CPLD
2F10 HBHUX_CPLD
2D10 HBLVX_CPLD
2E10 HBHWX_CPLD
1D10 HBLUXIN
2E10 HBHVXIN
1D10 HBLWXIN
2G10 NFLTBLX_CPLD
1J9 NFSENX_CPLD
2G10 FSLSX_CPLD
8E11 IUX_A
8E11 IWX_A
8D12 HWVERX
<TODO>:HWVER
GSENLX_CPLD 2G10
DISHDFWX_CPLD 2C10
HBLUX_CPLD 2D10
HBHVX_CPLD 2F10
HBLWX_CPLD 2D10
HBHUXIN 2E10
HBLVXIN 1D10
HBHWXIN 2E10
NFLTBHX_CPLD 2H10
DISHVASCX_CPLD 2B10
FSHSX_CPLD 2G10
IVX_A 8E11
HVUX 8D12
ERNI SMC Male straight
2J10 NOVPX_CPLD
R1411 47 DISCHGX
4C7 DRVX_CSL1_P10_0
4C7 DRVX_CSH1_P10_4
4C7 DRVX_CLK1_P10_2
4C7 DRVX_MRST1_P10_1
4C7 DRVX_MTSR1_P10_3
R1413 47 RDYL CSLX
R1414 47 RDYH CSHX
R1415 47 ENDRVHX CLKX
R1416 47 MRSTX MRSTX
R1417 47 ENDRVLX MTSRX
P10_2:ISG上桥驱动芯片EN
P10_3:ISG下桥驱动芯片EN
For SMT
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NO. Revision DSN. Date Name Date
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CHK.
ADT.
Title: Internal interface
IMD
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NO.	Revision	DSN.	Date		Name	Date
				DSN.		
				DWG.		
				CHK.		
				ADT.		

Title: Chopper and Interlock

IMD

NO. <Doc>

SIZE Custom

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